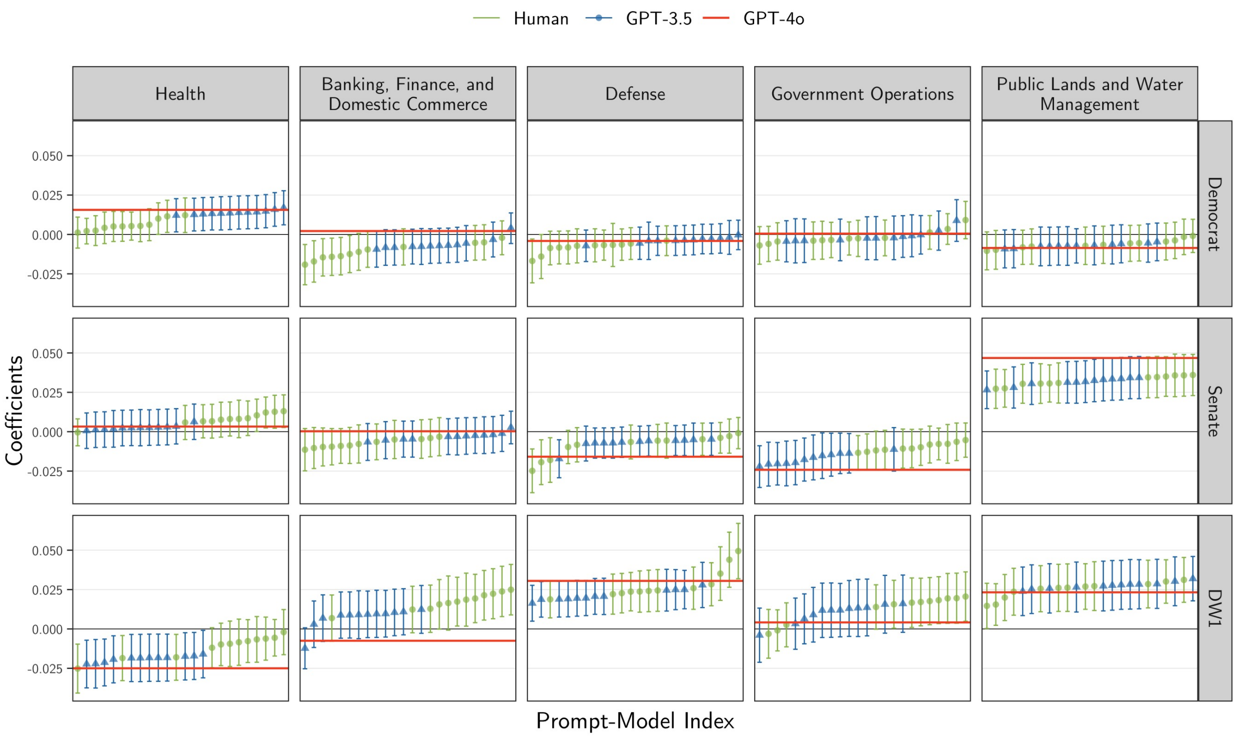
Results for Regressions with LHS Proxy

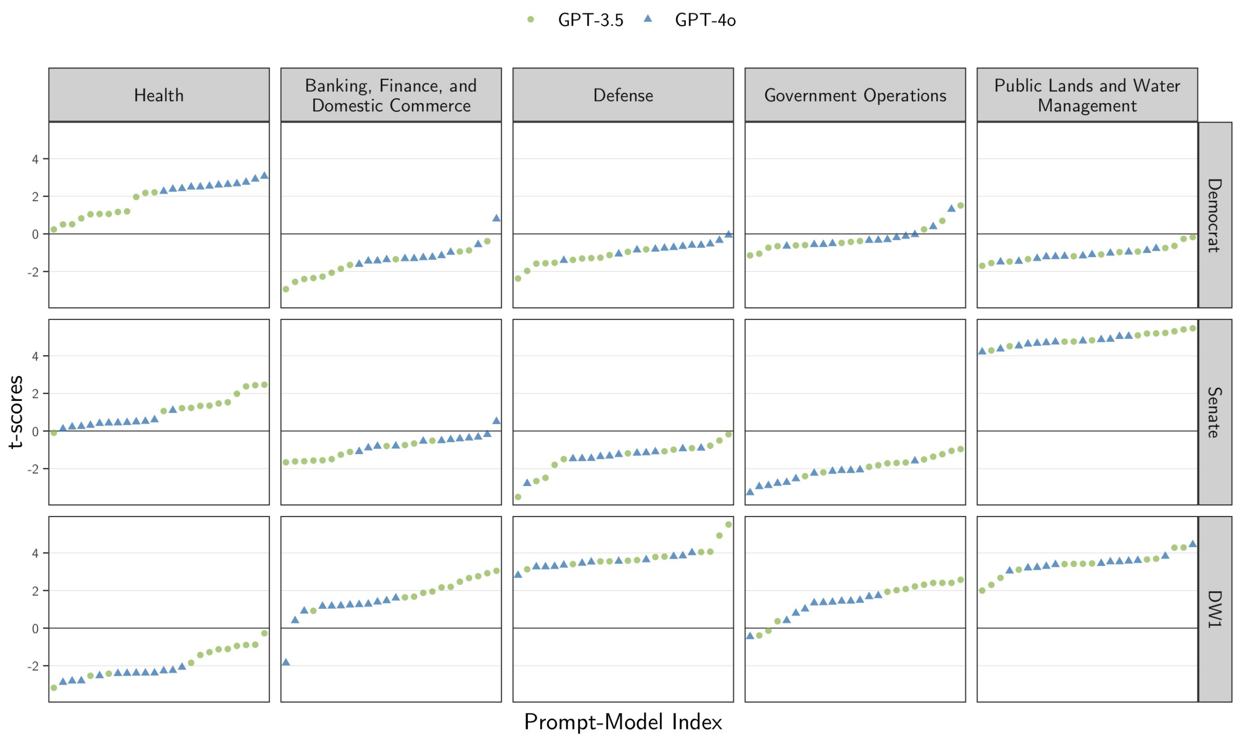
Sep 24, 2024

Figure 1: β vs. Prompt. Proxy on the LHS.



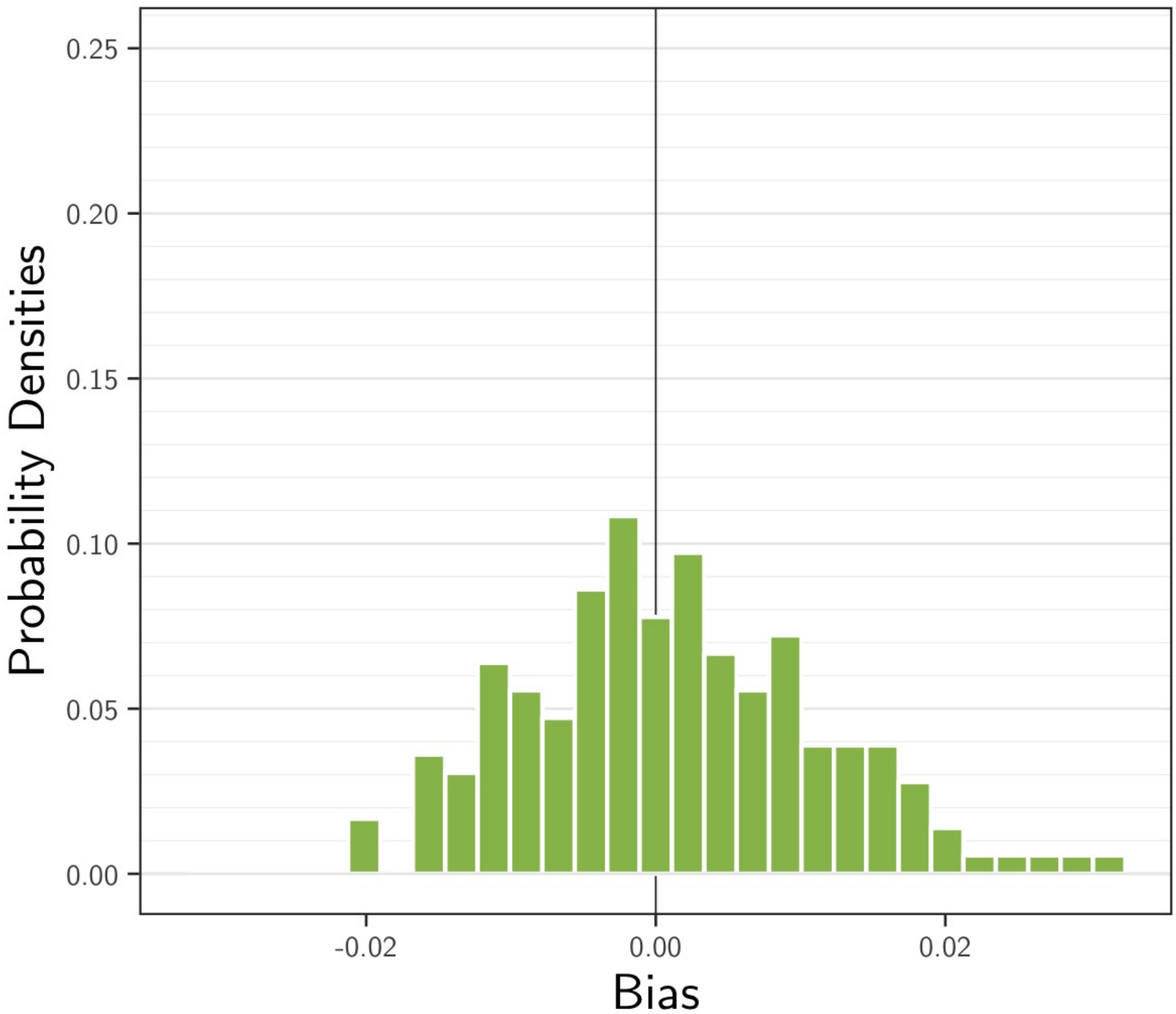
Note: The coefficients were obtained by fitting the model $Y_t = \alpha + \beta V$ using 10,000 bills, with 95% confidence intervals computed via robust standard errors. Here, Y_t are indicator for topic $t \in \{$Health; Banking, Finance, and Domestic Commerce; Defense; Government Operations; Public Lands and Water Management; Other$\}$ and $V \in \{\text{Democrat}; \text{Senate}; \text{DW1}\}$ are the covariates. The human-based coefficients $\beta^\star$ (red) were estimated using $Y_t^{\text{Human}} := 1\{Y^{\text{Human}} = t\}$, while LLM-based coefficients β^{LLM} (GPT-3.5 in green circles, GPT-4o in blue triangles) used $Y_{t,m,p}^{\text{LLM}} := 1\{Y_{m,p}^{\text{LLM}} = t\}$, with $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denoting the LLM model and $p \in \{1, \ldots, 12\}$ representing the prompt. In each subplot, the bias estimates were sorted in ascending order for clarity.

Figure 2: t-scores of β vs. Prompt. Proxy on the LHS.



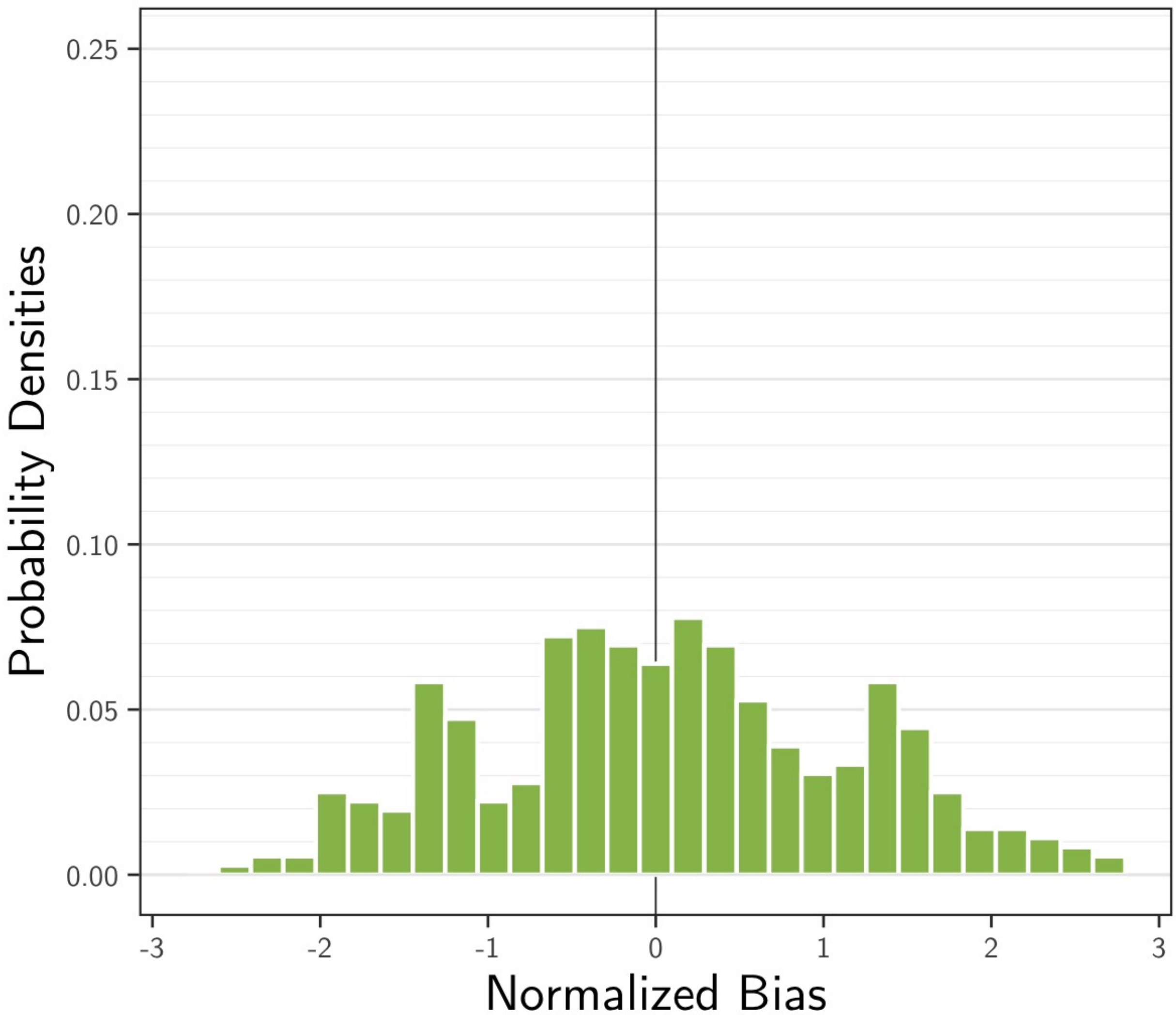
Note: The t-scores were obtained by fitting the model $Y_t = \alpha + \beta V$ using 10,000 bills, with 95% confidence intervals computed via robust standard errors. Here, Y_t are indicator for topic $t \in$ {Health; Banking, Finance, and Domestic Commerce; Defense; Government Operations; Public Lands and Water Management; Other} and $V \in \{\text{Democrat}; \text{Senate}; \text{DW1}\}$ are the covariates. The human-based coefficients $\beta^\star$ (red) were estimated using $Y_t^{\text{Human}} := 1\{Y^{\text{Human}} = t\}$, while LLM-based coefficients β^{LLM} (GPT-3.5 in green circles, GPT-4o in blue triangles) used $Y_{t,m,p}^{\text{LLM}} := 1\{Y_{m,p}^{\text{LLM}} = t\}$, with $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denoting the LLM model and $p \in \{1, \ldots, 12\}$ representing the prompt. In each subplot, the bias estimates were sorted in ascending order for clarity.

Figure 3: Bias Density of β^{LLM}. Proxy on the LHS.



Note: The bias estimates were calculated as the difference between the LLM-based coefficient β^{LLM} and the human-based coefficient $\beta^\star$, using a regression of the form $Y_t = \alpha + \beta V$, where $t \in$ {Health; Banking, Finance, and Domestic Commerce; Defense; Government Operations; Public Lands and Water Management; Other} represents major topics and $V \in \{\text{Democrat}; \text{Senate}; \text{DW1}\}$ are the covariates. For $\beta^\star$, $Y_t^{\text{Human}} := 1\{Y^{\text{Human}} = t\}$, while for β^{LLM}, $Y_{t,m,p}^{\text{LLM}} := 1\{Y_{m,p}^{\text{LLM}} = t\}$, with $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denoting the LLM model and $p \in \{1, \ldots, 12\}$ representing the prompt. The human-based coefficient $\beta^\star$ was computed using the full 10,000 bills, whereas the LLM-based coefficient β^{LLM} was calculated as the average of $N = 1,000$ simulation runs, with each run's estimate based on a sample of 5,000 bills. These bias estimates are independent of the validation proportion η, though they were generated in the experiment run with $\eta = 5\%$.

Figure 4: Normalized Bias Density of β^{LLM}. Proxy on the LHS.



Note: The normalized bias estimates were calculated as the difference between the LLM-based coefficient β^{LLM} and the human-based coefficient $\beta^{\star}$, divided by the sample standard deviation computed across $N = 1000$ samples. All other variables are as defined in the previous figure. These bias estimates are independent of the validation proportion η, though they were generated in the experiment run with $\eta = 5\%$

Figure 5: Bias Density of β by Model and Proportion of Validation Samples. Proxy on the LHS.

Figure 6: Normalized Bias Density of β by Model and Proportion of Validation Samples. Proxy on the LHS.

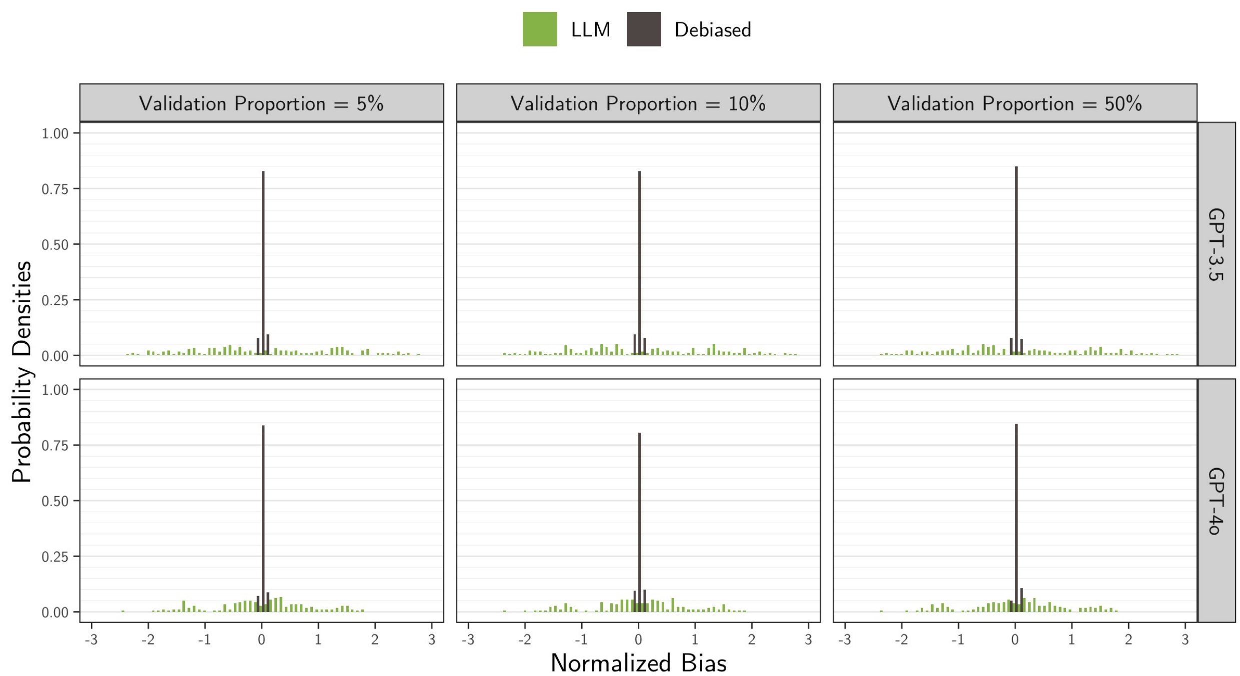


Figure 7: MSE Empirical CDF of β by Proportion of Validation Samples. Proxy on the LHS.

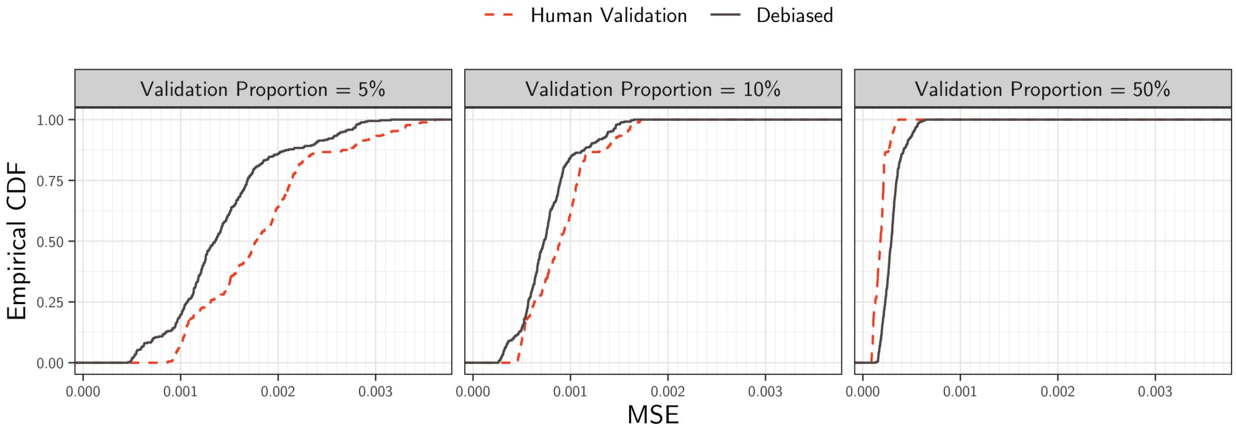


Figure 8: MSE Empirical CDF of β by Model and Proportion of Validation Samples. Proxy on the LHS.

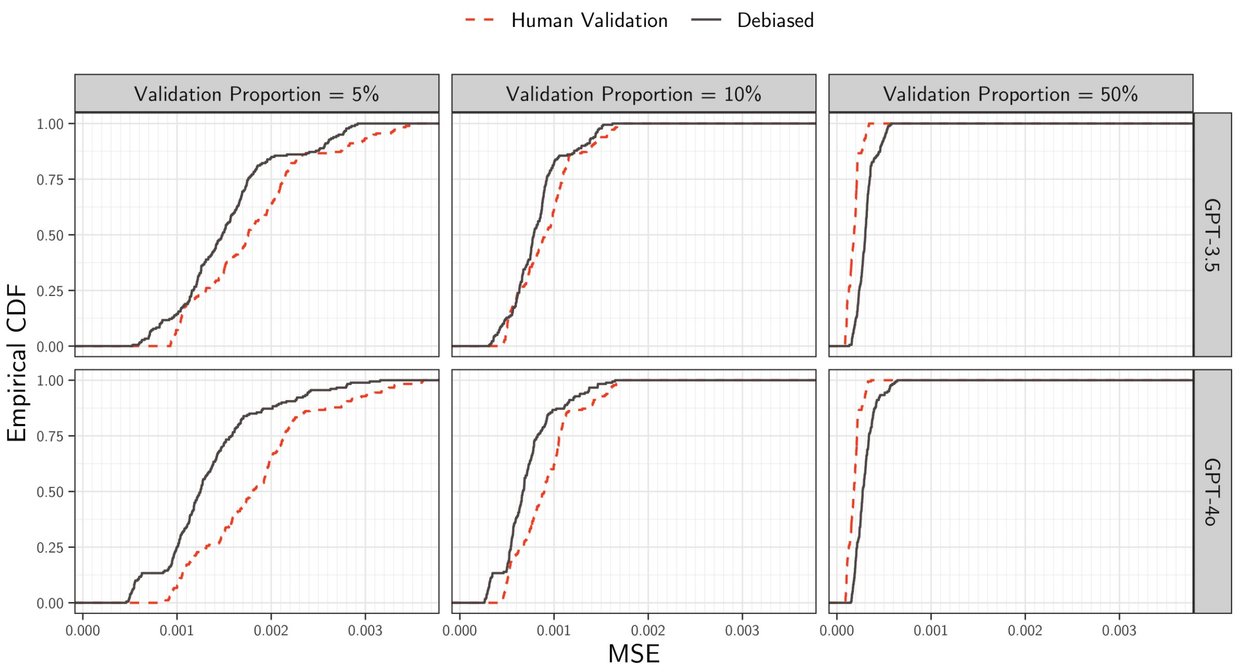


Table 1: Bias Summary Statistics of β by Proportion of Validation Samples. Proxy on the LHS: $Y = \alpha + \beta V$.

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.010	0.001	-0.015	0.018
5%	Human Validation	0.000	0.001	0.000	-0.002	0.002
5%	Debiased	0.000	0.001	0.000	-0.002	0.002
10%	LLM	0.001	0.010	0.000	-0.015	0.018
10%	Human Validation	0.000	0.001	0.000	-0.002	0.001
10%	Debiased	0.000	0.001	0.000	-0.002	0.001
25%	LLM	0.001	0.010	0.000	-0.014	0.018
25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
25%	Debiased	0.000	0.001	0.000	-0.001	0.001
50%	LLM	0.001	0.010	0.000	-0.014	0.019
50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
50%	Debiased	0.000	0.001	0.000	-0.001	0.001

Table 2: Normalized Bias Summary Statistics of β by Proportion of Validation Samples. Proxy on the LHS: $Y = \alpha + \beta V$.

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.071	1.099	0.067	-1.692	1.856
5%	Human Validation	0.000	0.034	0.001	-0.055	0.054
5%	Debiased	0.003	0.033	0.001	-0.055	0.061
10%	LLM	0.069	1.094	0.040	-1.722	1.869
10%	Human Validation	-0.002	0.030	-0.001	-0.055	0.048
10%	Debiased	-0.002	0.033	-0.002	-0.060	0.049
25%	LLM	0.073	1.099	0.023	-1.711	1.854
25%	Human Validation	0.002	0.031	0.001	-0.050	0.051
25%	Debiased	0.001	0.033	0.001	-0.049	0.057
50%	LLM	0.076	1.096	0.047	-1.654	1.875
50%	Human Validation	0.004	0.032	0.003	-0.051	0.056
50%	Debiased	0.000	0.031	-0.001	-0.050	0.053

Table 3: Bias Summary Statistics by Model and Proportion of Validation Samples for β. Proxy on the LHS: $Y = \alpha + \beta V$.

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.002	0.012	0.000	-0.016	0.023
GPT-3.5	5%	Human Validation	0.000	0.001	0.000	-0.002	0.003
GPT-3.5	5%	Debiased	0.000	0.001	0.000	-0.002	0.002
GPT-3.5	10%	LLM	0.002	0.012	0.000	-0.016	0.023
GPT-3.5	10%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-3.5	10%	Debiased	0.000	0.001	0.000	-0.002	0.001
GPT-3.5	25%	LLM	0.002	0.012	0.000	-0.016	0.023
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	25%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	50%	LLM	0.002	0.012	0.000	-0.016	0.022
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
GPT-3.5	50%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-4o	5%	LLM	0.001	0.008	0.001	-0.013	0.015
GPT-4o	5%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	5%	Debiased	0.000	0.001	0.000	-0.002	0.002
GPT-4o	10%	LLM	0.001	0.008	0.001	-0.014	0.014
GPT-4o	10%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-4o	10%	Debiased	0.000	0.001	0.000	-0.002	0.001
GPT-4o	25%	LLM	0.001	0.008	0.000	-0.013	0.014
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-4o	25%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-4o	50%	LLM	0.001	0.008	0.001	-0.013	0.015
GPT-4o	50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
GPT-4o	50%	Debiased	0.000	0.001	0.000	-0.001	0.001

Table 4: Normalized Bias Summary Statistics by Model and Proportion of Validation Samples for β. Proxy on the LHS: $Y = \alpha + \beta V$.

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.113	1.284	-0.023	-1.899	2.211
GPT-3.5	5%	Human Validation	0.001	0.034	0.003	-0.052	0.056
GPT-3.5	5%	Debiased	0.003	0.034	0.001	-0.055	0.066
GPT-3.5	10%	LLM	0.109	1.280	-0.005	-1.863	2.179
GPT-3.5	10%	Human Validation	-0.003	0.031	-0.001	-0.064	0.042
GPT-3.5	10%	Debiased	-0.003	0.032	-0.005	-0.053	0.049
GPT-3.5	25%	LLM	0.118	1.285	0.007	-1.837	2.264
GPT-3.5	25%	Human Validation	0.004	0.029	0.002	-0.046	0.050
GPT-3.5	25%	Debiased	0.002	0.031	0.002	-0.044	0.053
GPT-3.5	50%	LLM	0.116	1.282	-0.004	-1.864	2.172
GPT-3.5	50%	Human Validation	0.004	0.033	0.003	-0.052	0.060
GPT-3.5	50%	Debiased	0.000	0.030	0.000	-0.050	0.049
GPT-4o	5%	LLM	0.029	0.877	0.084	-1.411	1.514
GPT-4o	5%	Human Validation	-0.001	0.033	0.000	-0.055	0.051
GPT-4o	5%	Debiased	0.002	0.031	0.001	-0.055	0.054
GPT-4o	10%	LLM	0.028	0.871	0.059	-1.447	1.507
GPT-4o	10%	Human Validation	-0.001	0.030	-0.002	-0.053	0.049
GPT-4o	10%	Debiased	-0.001	0.034	0.000	-0.066	0.050
GPT-4o	25%	LLM	0.027	0.876	0.056	-1.422	1.463
GPT-4o	25%	Human Validation	0.001	0.032	-0.001	-0.050	0.052
GPT-4o	25%	Debiased	0.000	0.035	-0.001	-0.053	0.060
GPT-4o	50%	LLM	0.035	0.874	0.070	-1.441	1.510
GPT-4o	50%	Human Validation	0.004	0.031	0.003	-0.050	0.051
GPT-4o	50%	Debiased	0.000	0.031	-0.001	-0.045	0.059

Table 5: MSE Summary Statistics of β by Proportion of Validation Samples. Proxy on the LHS: $Y = \alpha + \beta V$.

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.000	0.000	0.000	0.000	0.000
5%	Human Validation	0.002	0.001	0.002	0.001	0.003
5%	Debiased	0.001	0.001	0.001	0.001	0.003
10%	LLM	0.000	0.000	0.000	0.000	0.000
10%	Human Validation	0.001	0.000	0.001	0.000	0.002
10%	Debiased	0.001	0.000	0.001	0.000	0.001
25%	LLM	0.000	0.000	0.000	0.000	0.000
25%	Human Validation	0.000	0.000	0.000	0.000	0.001
25%	Debiased	0.000	0.000	0.000	0.000	0.001
50%	LLM	0.000	0.000	0.000	0.000	0.000
50%	Human Validation	0.000	0.000	0.000	0.000	0.000
50%	Debiased	0.000	0.000	0.000	0.000	0.001

Table 6: MSE Summary Statistics by Model and Proportion of Validation Samples for β. Proxy on the LHS: $Y = \alpha + \beta V$.

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	5%	Human Validation	0.002	0.001	0.002	0.001	0.003
GPT-3.5	5%	Debiased	0.002	0.001	0.001	0.001	0.003
GPT-3.5	10%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	10%	Human Validation	0.001	0.000	0.001	0.000	0.002
GPT-3.5	10%	Debiased	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	25%	Human Validation	0.000	0.000	0.000	0.000	0.001
GPT-3.5	25%	Debiased	0.000	0.000	0.000	0.000	0.001
GPT-3.5	50%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-3.5	50%	Debiased	0.000	0.000	0.000	0.000	0.001
GPT-4o	5%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	5%	Human Validation	0.002	0.001	0.002	0.001	0.003
GPT-4o	5%	Debiased	0.001	0.001	0.001	0.001	0.002
GPT-4o	10%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	10%	Human Validation	0.001	0.000	0.001	0.000	0.002
GPT-4o	10%	Debiased	0.001	0.000	0.001	0.000	0.001
GPT-4o	25%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	25%	Human Validation	0.000	0.000	0.000	0.000	0.001
GPT-4o	25%	Debiased	0.000	0.000	0.000	0.000	0.001
GPT-4o	50%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	50%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-4o	50%	Debiased	0.000	0.000	0.000	0.000	0.001

Table 7: Coverage Summary Statistics of β by Proportion of Validation Samples. Proxy on the LHS: $Y = \alpha + \beta V$.

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.811	0.163	0.889	0.461	0.951
5%	Human Validation	0.946	0.008	0.946	0.932	0.959
5%	Debiased	0.928	0.012	0.929	0.904	0.945
10%	LLM	0.810	0.165	0.889	0.460	0.951
10%	Human Validation	0.948	0.007	0.949	0.936	0.959
10%	Debiased	0.940	0.008	0.941	0.926	0.952
25%	LLM	0.809	0.164	0.887	0.465	0.952
25%	Human Validation	0.949	0.007	0.949	0.936	0.960
25%	Debiased	0.946	0.007	0.946	0.934	0.957
50%	LLM	0.810	0.164	0.887	0.463	0.950
50%	Human Validation	0.950	0.007	0.950	0.939	0.961
50%	Debiased	0.948	0.008	0.948	0.935	0.959

Table 8: Coverage Summary Statistics by Model and Proportion of Validation Samples for β. Proxy on the LHS: $Y = \alpha + \beta V$.

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.761	0.187	0.819	0.381	0.945
GPT-3.5	5%	Human Validation	0.946	0.008	0.948	0.933	0.961
GPT-3.5	5%	Debiased	0.929	0.011	0.931	0.910	0.945
GPT-3.5	10%	LLM	0.759	0.189	0.812	0.383	0.949
GPT-3.5	10%	Human Validation	0.947	0.007	0.947	0.936	0.959
GPT-3.5	10%	Debiased	0.940	0.008	0.941	0.927	0.952
GPT-3.5	25%	LLM	0.760	0.188	0.815	0.362	0.945
GPT-3.5	25%	Human Validation	0.949	0.007	0.949	0.936	0.958
GPT-3.5	25%	Debiased	0.946	0.007	0.946	0.934	0.957
GPT-3.5	50%	LLM	0.760	0.190	0.815	0.369	0.950
GPT-3.5	50%	Human Validation	0.951	0.007	0.950	0.939	0.962
GPT-3.5	50%	Debiased	0.947	0.008	0.948	0.934	0.959
GPT-4o	5%	LLM	0.861	0.116	0.921	0.637	0.954
GPT-4o	5%	Human Validation	0.945	0.008	0.946	0.930	0.957
GPT-4o	5%	Debiased	0.926	0.014	0.927	0.902	0.945
GPT-4o	10%	LLM	0.860	0.117	0.919	0.642	0.952
GPT-4o	10%	Human Validation	0.949	0.007	0.949	0.936	0.959
GPT-4o	10%	Debiased	0.940	0.008	0.941	0.926	0.953
GPT-4o	25%	LLM	0.859	0.117	0.921	0.625	0.954
GPT-4o	25%	Human Validation	0.949	0.007	0.949	0.936	0.960
GPT-4o	25%	Debiased	0.946	0.007	0.946	0.934	0.958
GPT-4o	50%	LLM	0.860	0.115	0.919	0.635	0.950
GPT-4o	50%	Human Validation	0.950	0.007	0.949	0.939	0.961
GPT-4o	50%	Debiased	0.948	0.007	0.948	0.935	0.959